

Binal Dave

Working Student | Data Science, Analytics & Python

✉ binaldave23@gmail.com ☎ +49 160 1270960 📍 Dortmund, Germany 🔗 LinkedIn 🐙 GitHub

PROFILE

I have two years of professional experience working with business data and Python, and I'm now specialising further in Data Science through my M.Sc. at TU Dortmund. I've worked on real-world ML problems as well as end-to-end analytical applications, and I'm particularly interested in using data to generate insights that support product and business decisions.

PROFESSIONAL EXPERIENCE

ERP / Software Developer - Python, Data & Automation

01/2024 – 02/2026 | Ahmedabad, India

Silent Infotech Inc.

- Developed a Google Sheets-style project-data application for a BIM/VDC services company, translating project scopes, service structures and operational requirements into structured digital workflows, views and reports.
- Developed Python-based Odoo applications, automated workflows, reports and dashboards across Manufacturing, Inventory, CRM, Sales and Accounting.
- Developed a machine-learning solution using historical CRM opportunity data to predict expected closing timelines, including data preparation, feature engineering and model evaluation to support sales forecasting and planning.
- Built REST API and webhook interfaces connecting Odoo with ClickUp, Bill.com, Clover and other external systems.
- Supported the migration of more than one million financial and operational records from NetSuite to Odoo, including data mapping, validation, reconciliation and quality checks.
- Diagnosed production incidents using application logs, API payloads and PostgreSQL queries and documented the implemented corrective solutions.
- Collaborated with international technical and non-technical stakeholders and received the Star Performance Award in April 2025.

PROJECTS

Predictive Maintenance Analytics Platform - GitHub, Python, scikit-learn, FastAPI, Streamlit, Docker

- Developed a predictive-maintenance analytics platform using 10,000 sensor observations, including 339 recorded equipment failures, to identify equipment failure risk from industrial sensor data.
- Performed data preprocessing, feature engineering, class-imbalance analysis, model comparison and classification-threshold optimisation.
- Achieved an F1 score of 0.882, precision of 0.949, recall of 0.824 and ROC-AUC of 0.979.
- Deployed the end-to-end ML pipeline through FastAPI and an interactive Streamlit application, with Docker-based containerisation.

Industrial Maintenance RAG Assistant - GitHub, Python, FastAPI, Streamlit, FastEmbed, BM25, Groq, Docker

- Developed an end-to-end industrial maintenance analytical application using Python, FastAPI and Streamlit, integrating document-processing pipelines, information retrieval and grounded LLM responses.
- Processed 1,336 pages from three technical manuals into 2,421 metadata-preserving chunks with 384-dimensional BGE embeddings.
- Combined semantic retrieval, BM25, reciprocal rank fusion and exact technical-code matching, achieving Hit Rate@5 of 1.00 and MRR of 0.833.
- Delivered the application through FastAPI and Streamlit, with five automated API tests and Docker Compose deployment.

TECHNICAL SKILLS

Data Science & Machine Learning

Machine learning, predictive modelling, feature engineering, model evaluation, classification, data preprocessing, pandas, NumPy, scikit-learn

Data Engineering & Databases

SQL, PostgreSQL, ETL, data pipelines, data transformation, data modelling, data validation, reconciliation, API integration

Tools and Deployment

Git, GitHub, Docker, Docker Compose, FastAPI, Pytest, JavaScript, XML, QWeb and OWL.js

Software Engineering

Python, FastAPI, REST APIs, Git, GitHub, Pytest, Docker, modular application development, testing, documentation

Analytical Applications

Streamlit, dashboards, KPI reporting, Matplotlib, Seaborn

AI & Knowledge Systems

RAG, embeddings, semantic search, BM25, reciprocal rank fusion, LLM API integration

EDUCATION

M.Sc. Data Science, TU Dortmund University	04/2026 – Present
B.E. Computer Engineering, Government Engineering College, Rajkot	08/2020 – 05/2024

LANGUAGES AND AVAILABILITY

English - C1 | German - A2, actively learning | Gujarati - Native
Available for 15–20 hours per week; open to mobile work and regular attendance in Hannover by arrangement.